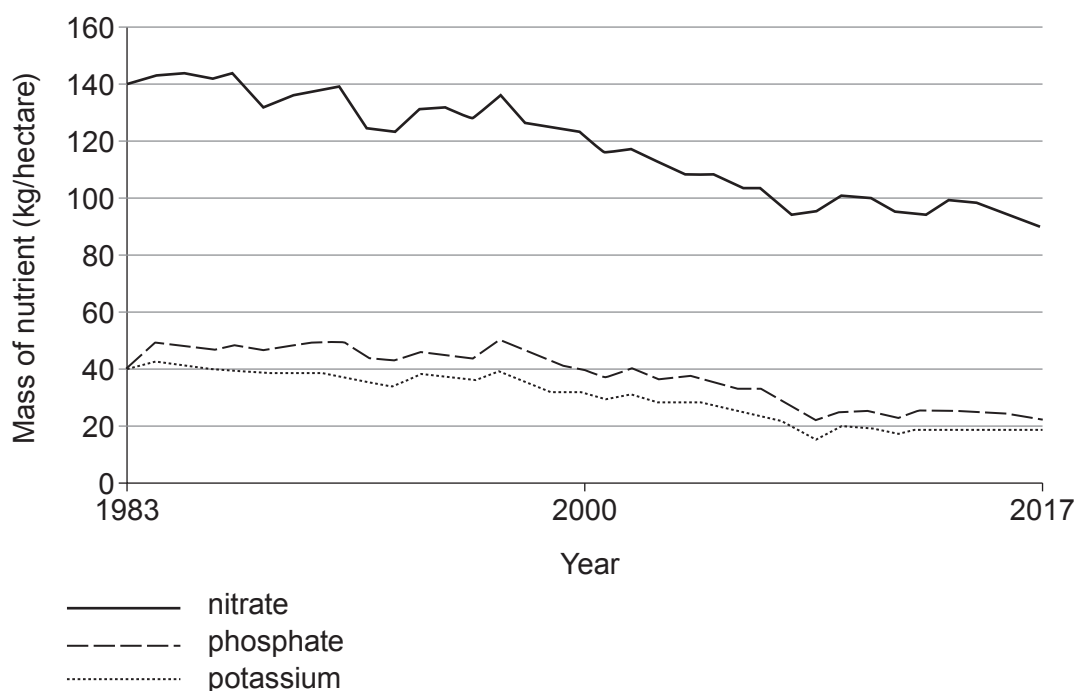


- (b) Healthy plants require a supply of nutrients. Many farmers use artificial fertilisers to supply these nutrients to the soil.

The graph shows the mass of different nutrients applied to agricultural land between 1983 and 2017.



Use the data from the graph to answer the following questions.

- (i) It is claimed that the use of nitrate fertilisers halved between 1983 and 2017. Use the graph to explain whether you agree.

[3]

.....

.....

.....



- (ii) The total land area of Wales is approximately 2 100 000 hectares.
60% of this land is used for farming.

I. Calculate the area of land used for farming in Wales. [1]

area = hectares

II. Calculate the total mass of phosphates used in farming in Wales in 1983. [2]

mass = kg

END OF PAPER

10





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1 4 He Helium 2																											
7 Li Lithium 3		9 Be Beryllium 4																									
23 Na Sodium 11		24 Mg Magnesium 12																									
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36								
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54								
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86								
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																							
Key																											

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number